

Postmenopausal status and early menopause as independent risk factors for cardiovascular disease: a meta-analysis

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ABSTRACT

Objective: Loss of ovarian function and subsequent deficiency of endogenous estrogens is suggested to enhance cardiovascular disease risk and related death after menopause. The aim was to obtain valid estimates of the cardiovascular disease risk associated with postmenopausal status and early menopause.

Design: A literature search of observational studies was performed using PubMed and EMBASE (1966 to May 1, 2004). Eighteen studies on postmenopausal status and age at menopause in relation to cardiovascular disease were selected. Six studies investigated menopausal status, nine studies investigated menopausal age, and three studied both. General variance-based methods were used to pool relative risk estimates and corresponding 95% confidence intervals. Stratification was performed for study design, type of menopause, outcome, and adjustment for age and smoking.

Results: The pooled relative risk estimate for postmenopausal versus premenopausal status and cardiovascular disease was 1.36 (95% CI, 1.15-1.60). In the stratified analysis, the pooled effect was 0.96 (95% CI, 0.77-1.21) after controlling for age and smoking. The pooled effect of bilateral oophorectomy on cardiovascular disease was 2.62 (95% CI, 2.05-3.35). For early menopause and cardiovascular disease, with the menopausal age category containing 50 years as a reference, the pooled relative risk estimate was 1.25 (95% CI, 1.15-1.35). In the stratified analysis, the pooled effect was 1.38 (95% CI, 1.21-1.58) after controlling for age and smoking. The pooled effect of bilateral oophorectomy on cardiovascular disease was 4.55 (95% CI, 2.56-8.01).

Conclusions: Overall, there was no convincing relationship between postmenopausal status and cardiovascular disease. However, there was a modest effect of early menopause on cardiovascular disease. The effect was more pronounced for women with an artificial menopause than for women with a natural menopause.

Key Words: Cardiovascular diseases – Menopause – Meta-analysis – Women.

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Most women in westernized societies die from cardiovascular disease (CVD). Although CVD is more frequent in men than in women at any given age, women tend to outlive men, and the absolute number of women dying from CVD is equal or even higher.¹ Premenopausal women seem to be protected against cardiovascular morbidity and mortality in comparison

TABLE 1. *Included studies in the meta-analysis*

Reference	Type of study	Participants	Menopause
Parrish et al ⁹	Cohort (retrospective)	Cases: 80 women with bilateral oophorectomy before time of normal menopause or before age of 50 y (autopsy records) Controls: 80 age-matched cancer patients (only primary carcinomas of the specific organ or organ system) and accident victims (autopsy records)	Women with surgical menopause compared with premenopausal women (reference) Age at surgical menopause of age ≤ 35 y compared with age at surgical menopause of age 46-50 (reference)
Manchester et al ¹⁰	Cohort (retrospective)	Cases: 20 women with bilateral oophorectomy at least 5 y previously with clinical symptoms of CHD Controls: 65 age-matched premenopausal women with clinical symptoms of CHD	Women with surgical menopause (bilateral oophorectomy) compared with premenopausal women (reference)
Gordon et al ¹¹	Cohort (prospective)	2,873 women from the Framingham Study. In total there were 28,670 person years and 70 cases	Women with natural or surgical (bilateral oophorectomy or/and hysterectomy) menopause compared with premenopausal women (reference)
Colditz et al ¹²	Cohort (prospective)	116,258 women from Nurses' Health Study. There were 659,034 person years and 269 cases	Postmenopausal women compared with premenopausal women (reference) Type of menopause: natural and surgical (bilateral oophorectomy and hysterectomy)
van der Schouw et al ¹³	Cohort (prospective)	12,115 postmenopausal women from the DOM Study. 824 cases were observed in 202,717 person years	Age at menopause ≤ 40 y compared with age at menopause 50 y (reference) Type of menopause: all causes, natural and surgical (bilateral oophorectomy or/and hysterectomy)
Jacobsen et al ¹⁴	Cohort (prospective)	19,309 postmenopausal women from the screening program for breast cancer, Norway. 2,767 cases were observed in 386,400 person years	Age at natural menopause of ≤ 43 y compared with age at menopause of 50-52 y (reference)
Jacobsen et al ¹⁵	Cohort (prospective)	5,279 postmenopausal women from the California Seventh-Day Adventists Study. 308 cases were observed in 55,650 person years	Age at natural menopause of 35-40 years compared with age at menopause of 49-51 y (reference)
Cooper and Sandler ¹⁶	Cohort (prospective)	3,191 postmenopausal women from NHANES I. 345 cases were observed	Age at natural menopause of ≤ 40 y compared with age at menopause of >50 y (reference).
Cooper et al ¹⁷	Cohort (prospective)	867 postmenopausal women from the Menstruation and Reproductive History Study. 45 cases were observed in 44,899 person years	Age at natural menopause of 28-45 y compared with age at menopause of ≥ 51 y
Hu et al ¹⁸	Cohort (prospective)	35,616 postmenopausal women from the Nurses' Health Study. 1,107 cases were observed in 354,326 person years	Age at natural menopause of <40 y compared with age at menopause of 50-54 y
de Kleijn et al ¹⁹	Cohort (prospective)	9,450 postmenopausal women from the screening program for breast cancer, The Netherlands. 2,439 cases were observed in 161,742 person years	Age at menopause ≤ 44 compared with age at menopause of 48-51 y Type of menopause: all causes, natural and surgical (bilateral oophorectomy or/and hysterectomy)
Rosenberg et al ²⁰	Case-control	Cases: 279 nurses hospitalized for MI Controls: 5,580 age-matched nurses without history of MI	Women with surgical (bilateral oophorectomy) or natural menopause compared with premenopausal women (reference) Age at menopause of <35 y compared with age at menopause 45-50 y (reference)

TABLE 1. (continued) *Included studies in the meta-analysis*

Outcome	Factors controlled for ^a	Relative risk RR, OR, HR (95% CI)
MI	Age, hypertension, diabetes, body weight, reproductive rates, hormone therapy	Menopausal status: RR 4.2 (1.8-9.6); early menopause: RR 8.5 (2.2-33.8) ^b
Coronary atherosclerosis documented with angiography	Age	Menopausal status: RR 0.7 (0.3-1.6) ^b
CHD	Age	Menopausal status. Natural menopause: RR 2.3 (0.8-6.5); surgical menopause: RR 4.8 (1.8-12.8) ^b
CHD (non-fatal MI + fatal CHD)	Age, smoking, hypertension, high cholesterol, diabetes, BMI, hormone therapy, family history	Menopausal status. Natural menopause: RR 1.1 (0.7-1.7); surgical menopause: RR 1.7 (0.9-8.6)
Mortality CHD and CVD	Age, smoking, hypertension, diabetes, BMI, fat distribution, hormone therapy, oral contraceptives, reproductive factors, history of CVD	Early menopause. CHD: HR 2.0 (1.3-3.0); CVD: HR 1.7 (1.3-2.2); early natural menopause and CVD: RR 1.7 (1.2-2.4); early surgical menopause and CVD: RR 3.8 (1.4-10.2)
Mortality IHD	Age, country, occupation	Early menopause: RR 1.1 (1.0-1.3)
Mortality IHD	Age, smoking, hypertension, diabetes, BMI, hormone therapy, physical activity, reproductive factors, education, dietary factors	Early menopause: RR 1.7 (1.0-2.8)
Mortality due to CHD, cerebrovascular disease, cancer, other causes	Age, smoking, BMI, hormone therapy, education, pregnancy history, race	Early menopause: CHD: RR 1.5 (0.7-3.4); cerebrovascular disease: RR 0.8 (0.2-3.4)
IHD	Age, smoking, diabetes, BMI, hormone therapy	Early menopause: RR 3.2 (1.1-9.8)
CHD and stroke	Age, smoking, hypertension, high cholesterol, diabetes, BMI, hormone therapy, family history, parity	Early menopause: CHD: RR 1.5 (0.9-2.5); stroke: RR 0.7 (0.3-2.0)
Mortality due to CVD, IHD, cerebrovascular diseases, other causes	Age, hypertension, diabetes, hormone therapy	Early menopause: CVD: HR 1.1 (0.9-1.3); IHD: HR 1.1 (0.8-1.4). Cerebrovascular disease: HR 1.0 (0.7-1.5)
MI	Age	Menopausal status. All causes: OR 1.6 (1.2-2.0); natural menopause: OR 0.9 (0.6-1.3); surgical menopause: OR 2.9 (2.1-4.0). Age at menopause. All causes: OR 7.4 (3.9-14.1); natural menopause: OR 5.0 (0.6-43.2); surgical menopause: OR 4.1 (1.8-9.3) ^b

(Continued on next page)

TABLE 1. (continued)

Reference	Type of study	Participants	Menopause
Rosenberg et al ²¹	Case-control	Cases: 255 women admitted to hospital for acute MI Controls: 802 age- and residence-matched women admitted to hospital for other disorders unrelated to oral contraceptive use	Postmenopausal women compared with premenopausal women (reference); causes of menopause not specified
Palmer et al ²²	Case-control	Cases: 858 women admitted to hospital for the first non-fatal MI Controls: 858 age- and residence- matched women without history of MI	Age at natural menopause of <45 y compared with age at menopause ≥50 y (reference)
Fioretti et al ²³	Case-control	Cases: 429 women with first episode of non-fatal acute MI, admitted to 30 coronary care units Controls: 863 women admitted to the same network of hospitals for acute diseases other than cardiovascular, neoplastic, or hormone-related conditions	Postmenopausal women compared with premenopausal women (reference) Age at menopause of <45 y compared with age at menopause of 50-52 y (reference). All causes of menopause, but not specified
Blumel et al ⁸	Case-control	Cases: 34 women with acute myocardial ischemia admitted to hospital	Postmenopausal women compared with premenopausal women (reference). All causes of menopause, but not specified
Witteman et al ²⁴	Cross-sectional	Controls: 102 age-matched healthy women. 613 women without history of CVD	Women with natural or surgical (bilateral oophorectomy) menopause compared with premenopausal women (reference)
Joakimsen et al ²⁵	Cross-sectional	2,588 postmenopausal women	Age at menopause of 35-48 y compared with age at menopause of 49-51 y (reference). All causes of menopause, but not specified

MI, myocardial infarction; CHD, coronary heart disease; IHD, ischemic heart disease; CVD, cardiovascular disease; RR, risk or rate ratio; OR, odds ratio; HR, hazard ratio; 95% CI, 95% confidence interval.

^a Includes variables used in exclusion criteria, for matching, or in multivariate analysis.

^b Own crude estimation calculated from data provided in article.

with men of similar age and postmenopausal women. Loss of ovarian function and subsequent deficiency of endogenous estrogens is believed to promote CVD and related death after menopause.¹

These observations lead to the view that the estrogen deficit accompanying menopause plays a dominant role in the increased CVD risk and that this risk might be reversed by pharmacological interventions

FIG. 1. Postmenopausal status and cardiovascular disease risk, overall analysis. Premenopausal status is the reference.

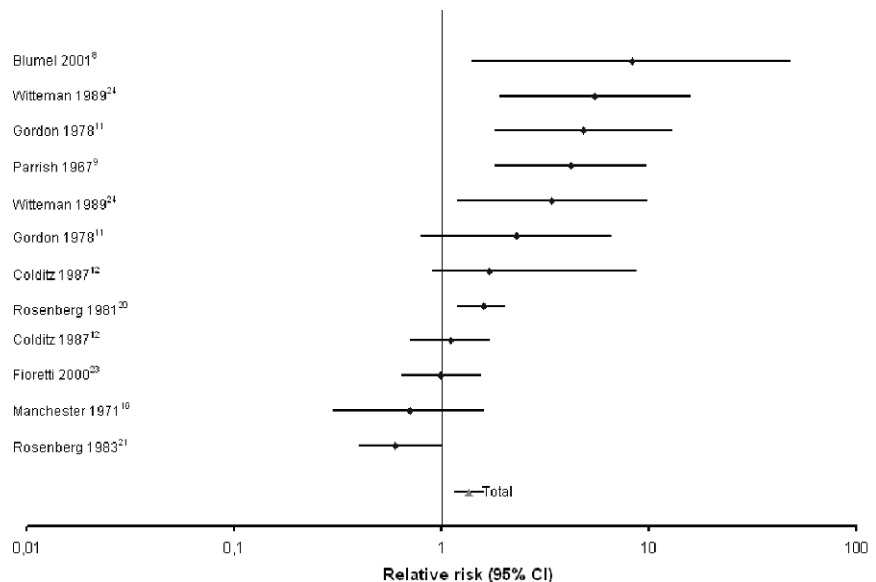


TABLE 1. (continued)

Outcome	Factors controlled for ^a	Relative risk RR, OR, HR (95% CI)
First non-fatal MI	Age, smoking, hypertension, high cholesterol, diabetes, obesity, hormone therapy, oral contraceptives, family history, reproductive factors, dietary factors	Menopausal status: OR 0.6 (0.4-1.0)
First non-fatal MI	Age, smoking, hypertension, high cholesterol, diabetes, BMI, hormone therapy, oral contraceptives, physical activity, family history, reproductive and dietary factors	Age at menopause: OR 2.1 (1.3-3.2)
Non-fatal acute MI	Age, smoking, hypertension, high cholesterol, diabetes, BMI, hormone therapy, education	Menopausal status: OR 1.0 (0.6-1.5); age at menopause: OR 0.7 (0.5-1.2)
Acute coronary syndromes	Age, smoking, hypertension, high cholesterol, diabetes, obesity, hormone therapy, family history	Menopausal status: OR 8.3 (1.4-47.5)
Calcified deposits in the abdominal aorta	Age, other risk indicators, but not specified	Menopausal status. Natural menopause: OR 3.4 (1.2-9.7); surgical menopause OR 5.5 (1.9-15.8)
Carotid atherosclerosis	Age, smoking, hormone therapy, education, parity, BMI	Age at menopause: OR 1.2 (1.0-1.5)

with exogenous hormones. Indeed, findings in observational studies seemed to suggest that hormone therapy may protect against increased postmenopausal cardiovascular disease risk.²⁻⁴ In contrast, recently

finished clinical trials on hormone therapy and cardiovascular disease did not corroborate the findings of the observational studies.⁵⁻⁷ In the Heart and Estrogen/Progestin Replacement Study (HERS), after

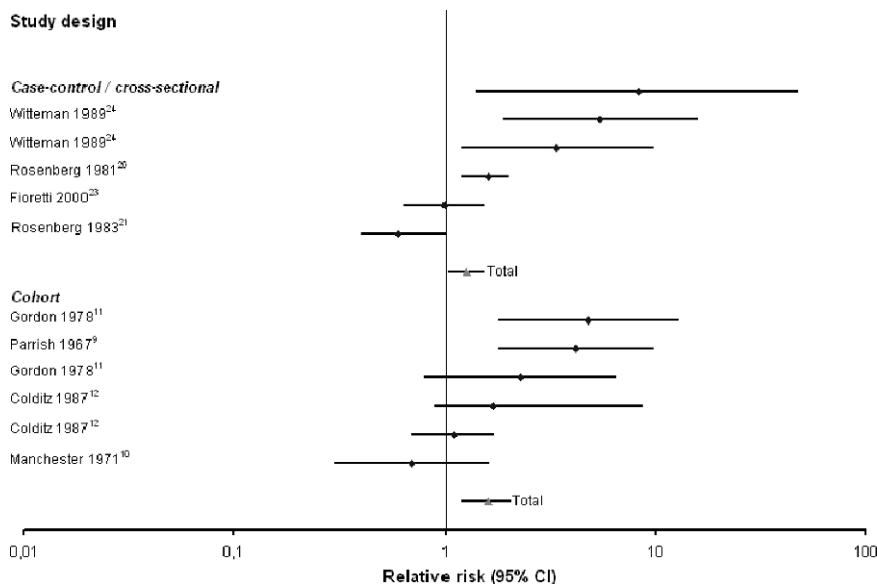
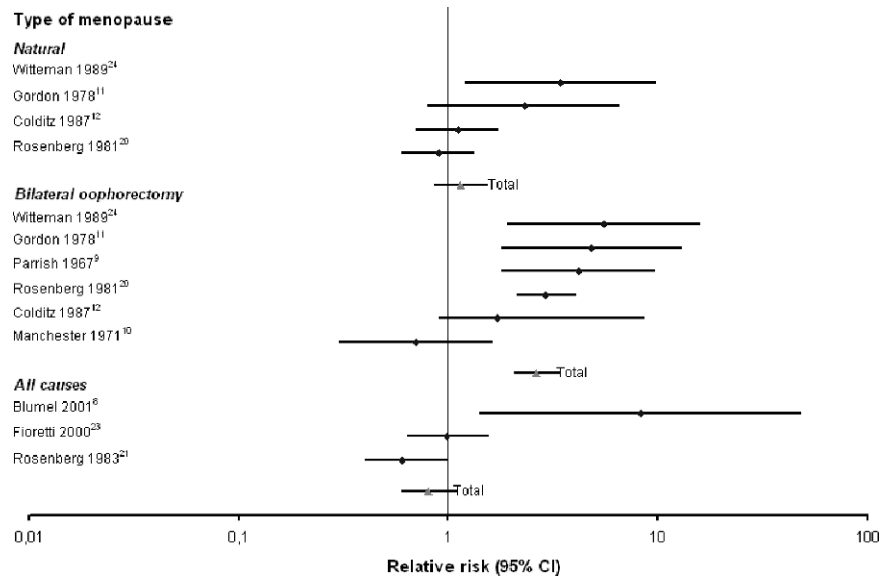


FIG. 2. Postmenopausal status and cardiovascular disease risk, stratification on study design. Premenopausal status is the reference.

FIG. 3. Postmenopausal status and cardiovascular disease risk, stratification on type of menopause. Premenopausal status is the reference.



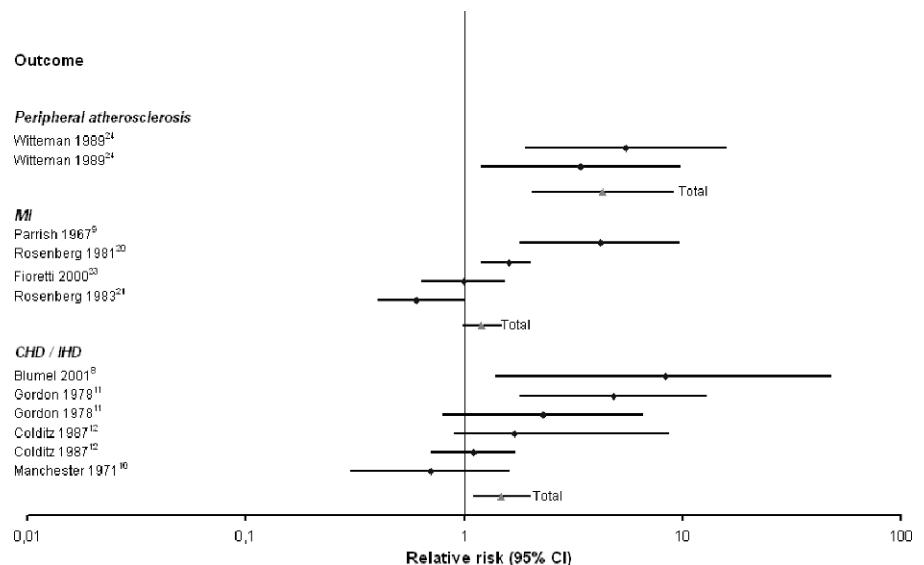
4.1 and 6.8 years of follow-up, hormone therapy did not reduce the risk of cardiovascular events in women with coronary heart disease.⁶ The estrogen/progestin arm of the Women's Health Initiative (WHI) trial was stopped early after an average 5.2-year follow-up among postmenopausal healthy women because women receiving hormone therapy had an increased risk of invasive breast cancer, along with evidence of some increase in coronary heart disease risk. The estrogen alone arm was stopped after a mean follow-up of 6.8 years because of an increased risk of stroke. However, the use of estrogen alone did not affect the incidence of coronary heart disease incidence.⁷

Together, these trials raise doubts on the hypothesized cardiovascular benefit of hormone therapy.

In view of these findings, one may doubt the inference from vital statistics of an increased risk of cardiovascular disease after the menopause. For decades, researchers have addressed the relationship between menopause and CVD. Because of the marked heterogeneity among the different studies, the question of whether menopause affects the incidence of CVD has still not been answered satisfactorily.

To obtain valid estimates of the cardiovascular disease risk with menopausal status and menopausal age, we summarized the evidence in available studies.

FIG. 4. Postmenopausal status and cardiovascular disease risk, stratification on outcome. Premenopausal status is the reference. MI, myocardial infarction; CHD, coronary heart disease; IHD, ischemic heart disease.



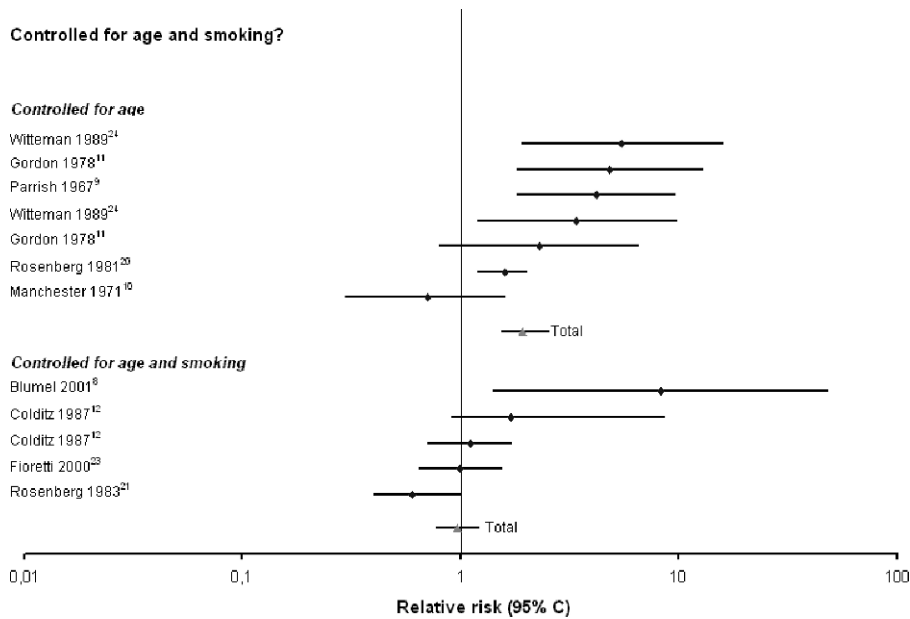


FIG. 5. Postmenopausal status and cardiovascular disease risk, stratification on controlling for age and smoking. Premenopausal status is the reference.

In addition, the extent to which differences in study design, type of menopause, outcome, and adjustment for age and smoking could explain discrepant results was examined.

METHODS

Data sources

Data for this meta-analysis were identified by searches in PubMed (1966 to 2004) using the terms “menopause” in combination with “cardiovascular disease” and “risk.” Articles about hormone therapy were excluded by the terms: “replacement,” “HRT,” “estrogen use,” “hormone use,” “exogenous,”

“treatment,” “medication,” or “therapy.” In addition, letters to the editor and overviews were excluded as well. Furthermore, the search was restricted to non-animal and non-laboratory studies. There was no restriction on language. EMBASE (1966 to 2004) was also searched, but no additional articles were selected.

Abstracts of 392 identified studies were printed out, and a further selection was made by three investigators (F.A., M.E.L.B., Y.T.S.) based on the inclusion criteria “menopause” (as determinant) and “cardiovascular disease” (as primary outcome). This selection was restricted to (1) original research; (2) etiologic research type as opposed to prognostic, diagnostic, and therapeutic research; and (3) the availability of a relative

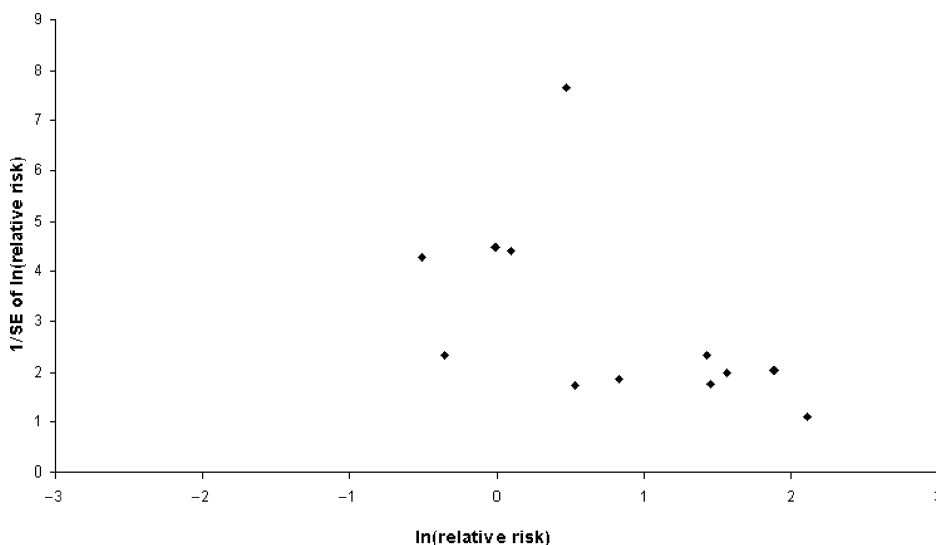
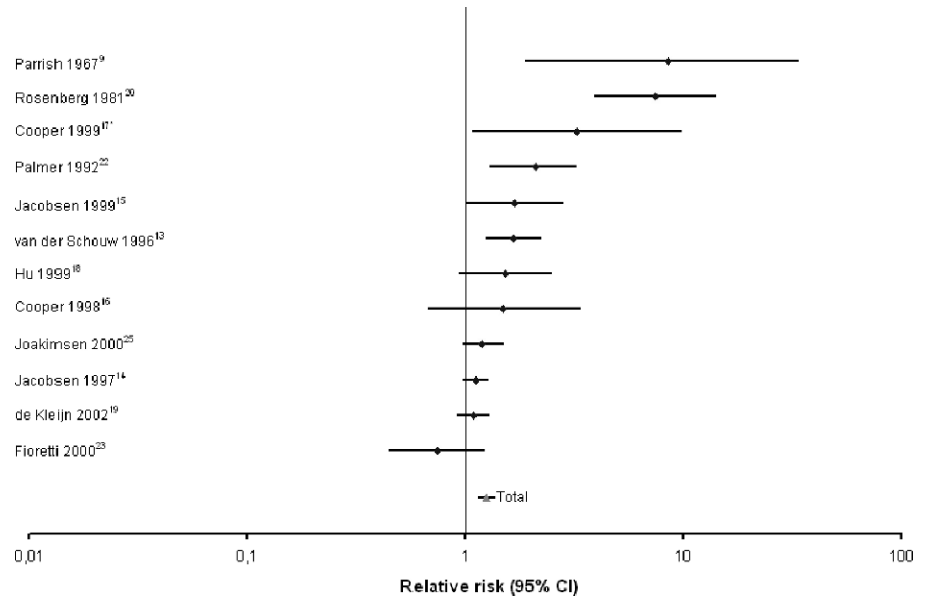


FIG. 6. Funnel plot of studies on postmenopausal status and cardiovascular disease. Kendall's tau-b correlation coefficient = -0.412, *P* value = 0.063.

FIG. 7. Early menopause and cardiovascular disease risk, overall analysis. Menopausal age category including 50 years as the reference. Menopausal age category with ≥ 51 years as the reference.



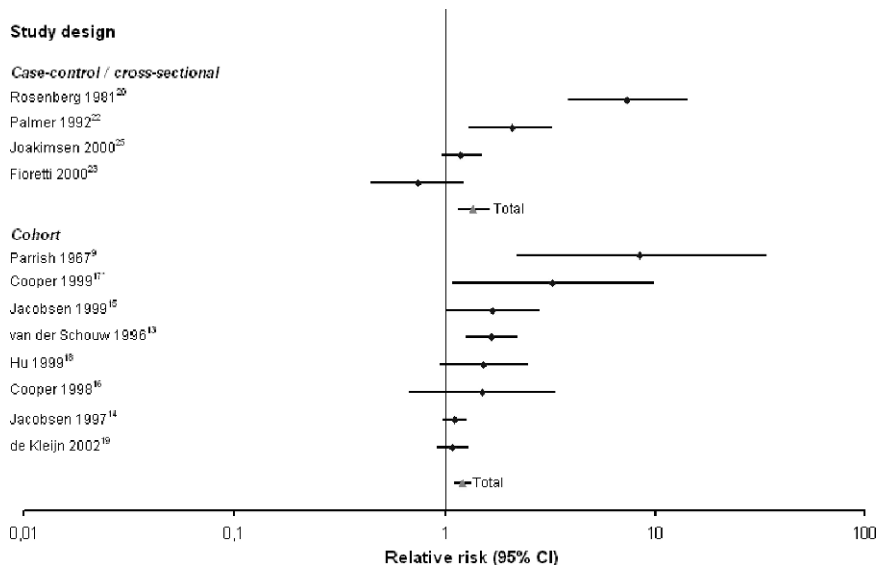
risk estimate and 95% confidence interval (95% CI) or the possibility to calculate this measure from raw data presented in the article. Relevant studies not identified by this strategy but referenced in the bibliographies of selected articles were also included. We focused on all publications referenced in PubMed and EMBASE but did not exclude commonly referenced and highly regarded older publications.

Study selection

From the 392 studies identified by the search strategy, 381 were excluded for various reasons. Seventy-three

reports were viewpoints, comments, or practical guidelines. Sixty-four publications were excluded because of lack of morbidity or mortality from CVD as an outcome measure. In 132 studies, the investigated determinants (endogenous estrogens, risk factors for CVD other than menopause, diet, and lifestyle factors) did not fit the objective of this article. There were also studies in which both outcome and determinant were the reason for exclusion (101 studies). Six studies were left out because study design and data were poorly described, and three studies were excluded because the used reference group did not meet our criteria. Furthermore,

FIG. 8. Early menopause and cardiovascular disease risk, stratification on study design. Menopausal age category including 50 years as the reference. Menopausal age category with ≥ 51 years as the reference.



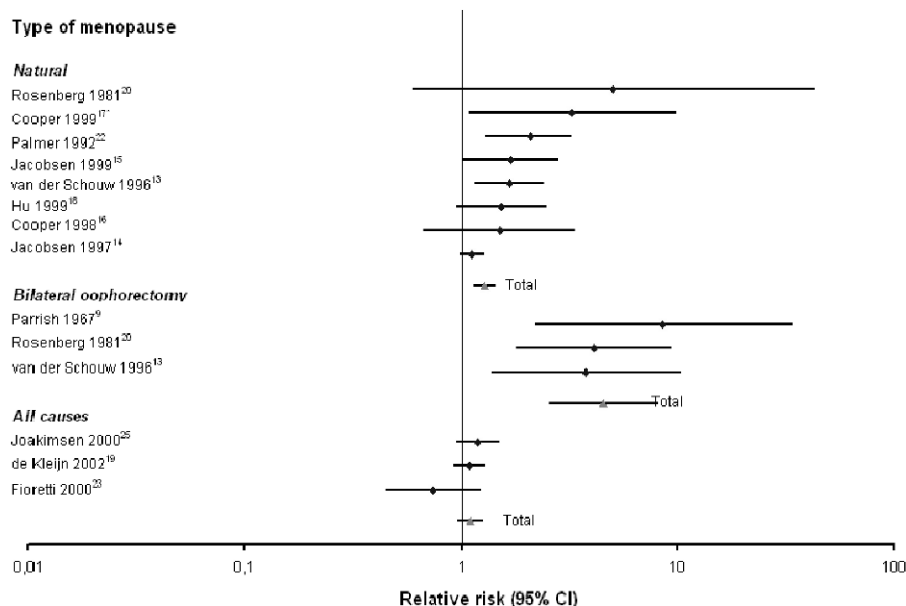


FIG. 9. Early menopause and cardiovascular disease risk, stratification on type of menopause. Menopausal age category including 50 years as the reference. Menopausal age category with ≥ 51 years as the reference.

one study relating risk factors to the probability of developing coronary heart disease in “health risk appraisal models” was excluded. If studies used the same cohort and research question for their analysis, the oldest study of these studies was excluded ($n = 1$).

The search resulted in 11 eligible studies. After scanning the bibliographies of these articles, another seven studies were selected. Finally, 18 articles were included in this review, 17 of which were written in English and 1 of which was in Spanish.⁸ Eleven studies were follow-up studies.⁹⁻¹⁹ A case-control design was used in five publications,^{8,20-23} and a cross-sectional design was

used in two publications.^{24,25} Six studies investigated menopausal status in relation to CVD, nine studies investigated menopausal age, and three studied both.

Data extraction

Data extraction was performed by one investigator (F.A.). A data extraction step was conducted for each study. Information was collected about study population, type of menopause, outcome, and the factors controlled for by means of exclusion criteria, matching, or multivariate analysis. If available, the adjusted relative risk estimates and corresponding 95% CI were

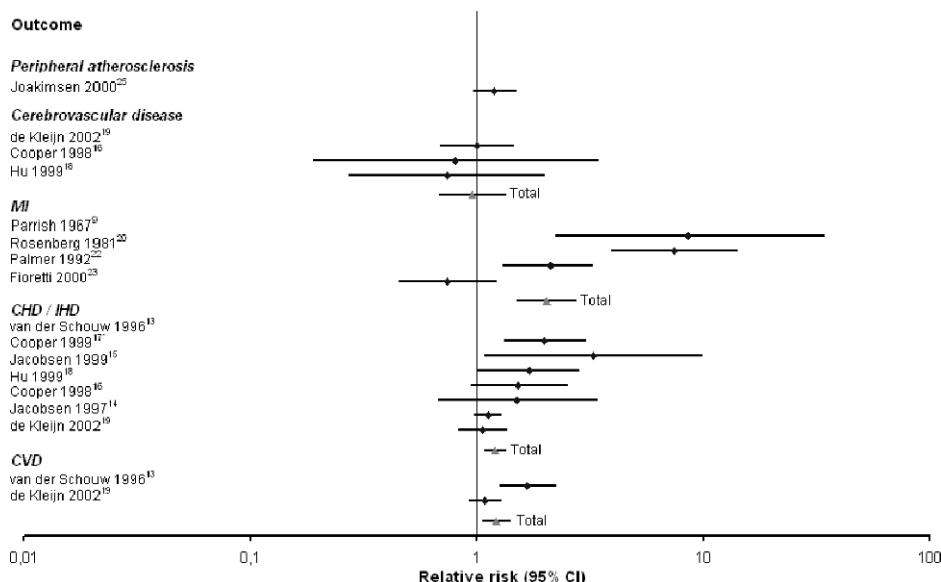
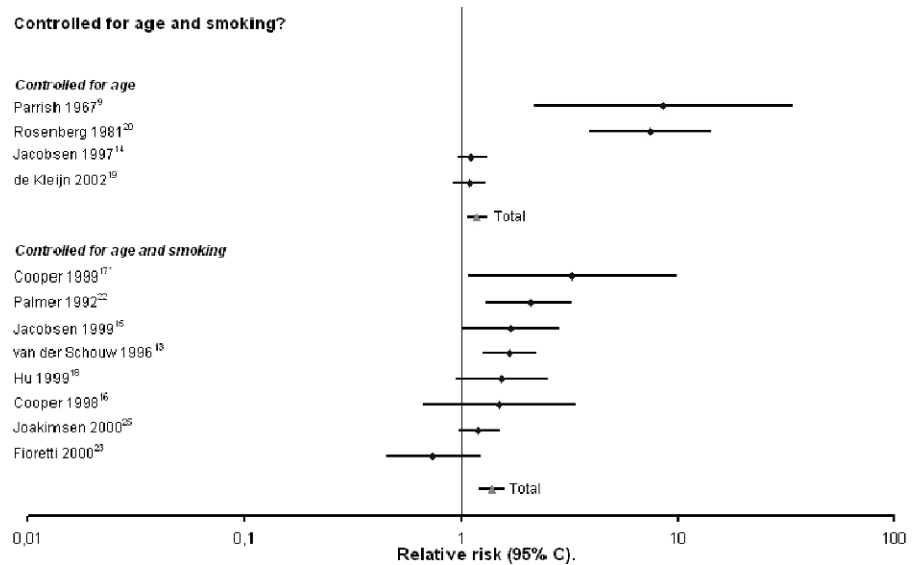


FIG. 10. Early menopause and cardiovascular disease risk, stratification on outcome. Menopausal age category including 50 years as the reference. Menopausal age category with ≥ 51 years as the reference. MI, myocardial infarction; CHD, coronary heart disease; IHD, ischemic heart disease; CVD, cardiovascular disease.

FIG. 11. Early menopause and cardiovascular disease risk, stratification on controlling for age and smoking. Menopausal age category including 50 years as the reference. Menopausal age category with ≥ 51 years as the reference.



copied from the original article. If no relative risk estimate was available, we calculated it from the raw data provided in the article using the statistical computer program Confidence Interval Analysis (CIA). In studies dealing with person time, a standardized rate ratio was calculated, in which a weighted average of the underlying stratum specific rate ratios was taken into account.

Data synthesis

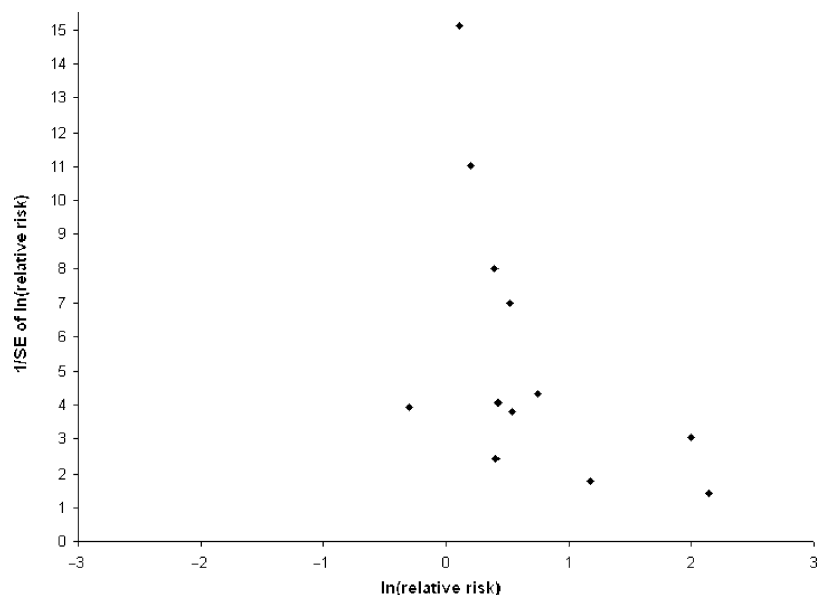
The studies were summarized according to their relative risk estimate and corresponding 95% CI. Testing for among-study heterogeneity was performed with the quantity I^2 , which describes the

percentage of total variation across studies that is the result of heterogeneity rather than change.^{26,27}

Pooling was performed separately for studies on postmenopausal status versus premenopausal status and for studies on early menopause. For the latter studies, the menopausal age category including 50 years was taken as a reference, because this is considered as the mean menopausal age. Exact category boundaries differed slightly across studies.

To calculate a pooled relative risk estimate and corresponding 95% CI, we used general variance-based methods weighting individual adjusted relative risk estimates by the inverse of their variance.^{28,29} Overall relative risk estimates presented in the articles, instead

FIG. 12. Funnel plot of studies on early menopause and cardiovascular disease. Kendall's tau-b correlation coefficient = -0.545 , P value = 0.014 .



of stratum-specific relative risk estimates (eg, separate estimates for natural and surgical menopause), were used to calculate the overall pooled effect. When authors only presented stratum-specific relative risk estimates without an overall estimation, these stratum-specific results were used under the condition that the populations in the different strata did not overlap.^{11,12,24}

Subgroup analyses were performed for the following predetermined variables: study design, type of menopause, outcome, and whether the authors controlled for the most important confounding variables of age and smoking. Studies were divided into two types of design: cohort studies, case-control and cross-sectional studies. Type of menopause was divided into three groups: all causes of menopause, bilateral oophorectomy, and natural cause. For outcome, studies were classified in the following subgroups: cardiovascular disease, coronary heart disease and ischemic heart disease, myocardial infarction, cerebrovascular disease, and peripheral atherosclerosis. Furthermore, we investigated whether authors controlled for age and smoking. Studies were divided into two classes: those controlling only for age and those that controlled for age and smoking.

Pooled relative risk estimates and corresponding 95% CIs were calculated for the different strata. Stratum-specific relative risk estimates were extracted from the articles and used for pooling. Therefore, the number of relative risk estimates used in the subgroup analyses could be larger than the number of relative risk estimates used in the overall analyses.

Finally, funnel plots were drawn, and Kendall's tau-b correlation coefficients for the correlation between effect size (logarithm of the relative risk) and study weight (1/standard error of the logarithm of the relative risk) were calculated to detect possible publication bias.

RESULTS

Included studies

To facilitate comparison among different studies, an overview of all selected studies is given in Table 1. The earliest publication dated from 1967⁹ and the most recent from 2002.¹⁹ Adjustment for confounding variables was poor in the older studies. Prospective studies including many subjects with a long follow-up and adjusting for confounding variables yielded the most valid results. The Framingham Study was the first prospective study on menopause and cardiovascular disease and was published in 1976.³⁰ The cohort had been followed since 1948 and included 2,873 woman aged 29 to 62 years. In 1978, a reanalysis of

the data was published.¹¹ After correction for age, an increase in coronary heart disease incidence after the menopause and a dramatic increase in the severity of the presenting disease was noted. The Nurses' Health Study was a large cohort established in 1976 and included 116,258 premenopausal women aged 30-55 years.¹² After correction for cardiovascular risk factors, no relationship was found between natural menopause (as opposed to premenopausal status) and coronary heart disease. There was a relationship between surgical menopause (as opposed to premenopausal status) and coronary heart disease, but this was still not statistically significant. In 1996, results were published from analyses of data for 12,115 women from the Doorlopend Onderzoek Morbiditeit/Mortaliteit (DOM) project who had been followed since 1974.¹³ After correction for age, it seemed that for each year's delay in menopause, the cardiovascular mortality risk decreased by 2%. Adjustment for other known cardiovascular risk factors did not alter the results. The adjusted risk of ischemic heart disease was increased in women who experienced early menopause. Another large prospective cohort of 19,309 postmenopausal women was published in 1997.¹⁴ A protective effect of late menopause for ischemic disease was found. However, no data were available on smoking, and the results were only adjusted for age, country, and occupation group. In a cohort of 5,279 Californian Seventh-Day Adventist women, similar results were found after a 13-year follow-up.¹⁵ In 1999, the effect of age at menopause on coronary heart disease was analyzed in the Nurses Health Study.¹⁸ An overall significant association between younger age at natural menopause and higher risk of CHD was observed. The most recent prospective follow-up study on age at menopause and cardiovascular disease was published in 2002.¹⁹ Although data on smoking were not available in this study, early menopause seemed to be related to cardiovascular disease.

Postmenopausal status

Nine studies investigated the relation between the postmenopausal status and cardiovascular disease (Fig. 1). In total, 12 relative risk estimates and 95% CIs were used for the overall meta-analysis. There was substantial among-study heterogeneity [$I^2 = 78\%$ (62%-87%)].

A comparison of the occurrence of CVD was made between postmenopausal status and premenopausal status, with premenopausal status as a reference. The relative risk estimates ranged from 0.6 (95% CI, 0.4-1.0)²¹ to 8.3 (95% CI, 1.4-47.5).⁸ The overall

pooled relative risk estimate for the relation between postmenopausal status, as opposed to premenopausal status, and CVD was 1.36 (95% CI, 1.15-1.60).

Figures 2 to 5 display results of the subgroup analyses. In the stratified analysis for type of menopause, the pooled effect of bilateral oophorectomy on CVD was 2.62 (95% CI, 2.05-3.35) and differed from the pooled effect of natural menopause on CVD, which was 1.14 (95% CI, 0.86-1.51) (Fig. 3). After controlling for confounding by age and smoking, the pooled effect of postmenopausal status on cardiovascular disease was 0.96 (0.77-1.21) (Fig. 5).

Figure 6 presents a funnel plot of the previously described studies. The lower left of the funnel plot, representing the non-significant risk estimates in small studies, showed a gap, and the plot had a rather asymmetrical distribution. The Kendall's tau-b correlation coefficient for the correlation between effect size (logarithm of the relative risk) and study weight (1/standard error of the logarithm of the relative risk) was -0.412 (P value = 0.063). These results point to the presence of publication bias.

Age at menopause

Twelve studies investigated age at menopause in relation to cardiovascular disease (Fig. 7). Twelve relative risk estimates and corresponding 95% CIs were used for the overall meta-analysis. There was substantial among-study heterogeneity [$I^2 = 82\%$ (69%-89%)].

We compared early menopause with the menopausal age category including 50 years (as the reference) with respect to CVD. The relative risk estimates varied from 0.7 (95% CI, 0.5-1.2)²³ to 8.4 (95% CI, 2.2-33.8).⁹ The pooled relative risk estimate for early menopause and CVD risk was 1.25 (95% CI, 1.15-1.35).

Results of subgroup analyses are shown in Figs. 8-11. In the stratified analysis for type of menopause, the pooled effect of bilateral oophorectomy on CVD was 4.55 (95% CI, 2.56-8.01) and differed from the pooled effect of natural menopause on CVD [1.27 (95% CI, 1.14-1.43)] (Fig. 9). After stratification for outcome, the pooled relative risk estimate for early menopause and myocardial infarction was 2.03 (95% CI, 1.51-2.73) and higher than the pooled effects in other strata (Fig. 10). After controlling for age and smoking, the pooled effect of early menopause and cardiovascular disease was 1.38 (95% CI, 1.21-1.58) (Fig. 11).

The funnel plot in Fig. 12 representing studies on age at menopause and CVD also showed an

asymmetrical distribution. The Kendall's tau-b correlation coefficient for the correlation between the effect size (logarithm of the relative risk) and study weight (1/standard error of the logarithm of the relative risk) was -0.545 (P value = 0.014). These results might be an indication of publication bias.

DISCUSSION

In this meta-analysis, we summarize the results of a large number of observational studies on the effect of menopause and menopausal age on cardiovascular disease. Overall, the data point to an increased risk of cardiovascular disease associated with postmenopausal status as opposed to premenopausal status and early menopause as opposed to the menopausal age category of 50 years. However, studies adjusting for age and smoking showed no relationship between postmenopausal status and cardiovascular disease.

One study¹⁷ used a menopausal age of 51 years and older as a reference. We investigated the influence of this study on the pooled estimates. When we excluded the article, the pooled estimates did not change materially.

The studies included in this meta-analysis were rather heterogeneous with respect to publication date, study size, study design, type of menopause, outcome, and the way potentially confounding factors were taken into account. Despite a significant test for heterogeneity, we decided to calculate a pooled estimate. This might be a problem because pooling of heterogeneous studies may lead to invalid pooled results. Because we cannot exclude the possibility that the pooled results are biased, we also presented the results of individual studies. We considered the results of individual studies compatible with the results of pooled estimates because the data from these individual studies also point to the overall conclusion that postmenopausal status and younger age at menopause are modestly related to cardiovascular disease. Furthermore, the results of individual studies on postmenopausal status and cardiovascular disease were, like the pooled results, also less convincing when adjusted for age and smoking.

In subgroup analyses we investigated the effect of sources of heterogeneity. We evaluated all studies with respect to adjustment for the cardiovascular risk factors age and smoking. In our opinion, these are the most important confounding variables. Smoking has been shown to decrease age at menopause³¹ and is a well-known risk factor for cardiovascular disease. Age is related to menopausal status and may also be

related to age at menopause because postmenopausal women are included in studies on age at menopause. Women with a higher age at menopause may be older at time of inclusion in the study than women with a lower age at menopause. After controlling for confounding by age and smoking, the relationship between postmenopausal status and cardiovascular disease disappeared. In contrast, the effect of early menopause was more pronounced after controlling for age and smoking. We cannot explain the conflicting findings. Residual confounding cannot be excluded because some studies categorized smoking as ever/never or current yes/no, and age was sometimes categorized in a limited number of categories as well. However, the study of van der Schouw et al¹³ shows that adjusting for smoking did not change the effect estimate for age at menopause; therefore, the importance of adjusting for smoking is not entirely clear yet. Studies on postmenopausal status were in general older than studies on age at menopause. It might therefore also be possible that other unobserved sources of heterogeneity in these studies played a role. Studies primarily investigating age at menopause were large, more recent, and more frequently of a prospective nature; thus, these studies were generally better designed.

We found a particularly higher risk of CVD among women who underwent bilateral oophorectomy compared with women who experienced natural menopause. This might be explained by the sudden removal of the ovaries at a young age and the subsequent sudden decline in hormone levels, whereas in naturally menopausal women, the decline in hormone levels occurs gradually and usually at a later age.

The excess risk for peripheral atherosclerosis seems remarkable when investigating menopausal status, but the estimates might be driven by the small number of relative risk estimates that could be included in these strata.

It is possible that non-significant results in studies with low power have not been published. The studies for postmenopausal status and age at menopause were rather asymmetrically distributed in the funnel plot, which is compatible with publication bias. We acknowledge that this might be a problem. However, the effect of publication bias on the results and conclusions of systematic reviews has been investigated. With the "trim and fill" method, missing studies were imputed and in this way bias in funnel plots could be investigated. The investigators found that missing studies changed the conclusion in less than 10% of the meta-analyses.³²

In general, recall of age at menopause might be a problem, especially for older women for whom more time since menopause has passed. The validity of self-reported age at menopause has been investigated, and age at menopause was accurately reported within 1 year by 69% of the subjects, on the basis of the gynecologist's medical records.³³ In another study, the accuracy of age at natural menopause evaluated during an interview was compared to the age at menopause obtained from the medical record in four age groups (<44, 45-49, 50-54, and >55 years). 48% correctly ranked their age at menopause in the same age group.³⁴ Furthermore, data were analyzed from the Menstrual and Reproductive History study, a study of women whose menstrual status had been recorded from when they were at university until natural menopause over a period extending from 1934 to 1974. In this study, original data were compared with answers to a questionnaire administered in 1979. Of the 49 subjects, 75% correctly recalled their age at menopause within 1 year.³⁵ The validity and reproducibility of self-reported age at menopause was also investigated in women followed-up in a breast cancer screening project. Of these women, 70% recalled their age at menopause correctly within 1 year.³⁶ Therefore, recall bias does not seem to be a very large problem. If it occurred in some studies, it would most likely have been non-differential, and the effect would have been biased toward the null, leading to an underestimation of the true effect.

There has been a marked discussion whether a sudden increase in coronary disease susceptibility of women after the menopause truly exists when judged by vital statistics. Depending on the type of statistical analysis (displaying crude disease rates compared with age-specific rates or male to female age-specific death rates or semi-logarithmic graphs), a sudden increase in coronary disease risk after menopause may or may not be observed. Therefore, it has been argued that increases in CVD risk after menopause are an artifact of the way vital statistics are presented, instead of a real decreased protection from CVD after menopause.^{37,38} Later, this viewpoint was refuted by researchers who concluded that, under other assumptions, the observed coronary heart disease mortality pattern in women in their study population is not incompatible with an effect of menopause.³⁹ All together, it has become apparent that vital statistics are not a source that provides easy answers.

Assuming that the causal mechanism underlying the effect of the menopause on CVD risk is the

postmenopausal estrogen decline, one might expect hormone therapy to decrease this risk. However, in previously described clinical trials on hormone therapy and cardiovascular disease, such as the WHI trial and HERS, no effect of combined estrogen plus progestin on CVD was found.⁵⁻⁷ Explanations of inconsistencies among the results of the literature summarized in this review and clinical trials on hormone therapy remain unclear. It has been suggested that the addition of progestin to estrogen is problematic, because progestins may inhibit the beneficial effect of estrogen.⁴⁰ Another explanation might be that used dosages, formulations of oral estrogens, and administration of hormone therapy do not mimic the normal physiology and metabolism of endogenous estrogens. One important aspect in comparing observational data and randomized studies is the age of patients and the time since menopause. Nearly all observational data are derived from women who started hormone therapy for postmenopausal symptoms; thus, therapy was started in the early postmenopausal period. In the randomized trials on hormone therapy, however, women typically no longer suffered from perimenopausal complaints and started therapy 10-20 years or even longer after menopause. The time when hormone therapy is started relative to menopause might be crucial in the effectiveness of hormone therapy in the vascular system.

CONCLUSION

Overall, there was no convincing relationship between postmenopausal status and cardiovascular disease. However, there was a modest effect of early menopause on cardiovascular disease. The effect was more pronounced for women with an artificial menopause than for women who underwent natural menopause.

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